

# Dr. B. C. Roy Engineering College, Durgapur

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## Power Electronics Applications

Battery Charging, UPS, and SMPS Basics

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# Outline

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Introduction

Battery Charging

UPS Basics

SMPS Basics

Comparison and Summary

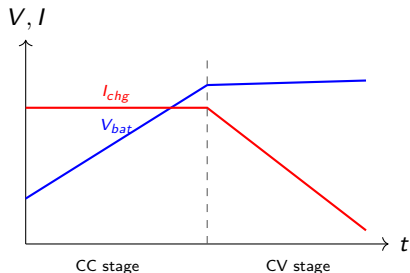
# Why These Applications Matter

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- Battery charging, UPS systems, and SMPS units all rest on the same foundation: efficient switching power conversion.
- **Battery charger** – converts and regulates power to safely replenish a battery.
- **UPS** – combines a charger, a battery, and an inverter to keep a load powered through a mains failure.
- **SMPS** – the general-purpose switching converter that both the above (and most modern electronics) are built from.
- Understanding one well makes the other two much easier to read.

# Charging Profile Basics – CC–CV

- Most rechargeable chemistries (Li-ion, SLA) are charged in two stages:
- **Stage 1 – Constant Current (CC):** the charger delivers a fixed current; battery voltage rises steadily. This is the “bulk” charging phase.
- **Stage 2 – Constant Voltage (CV):** once the target voltage is reached, it is held fixed; current tapers down as the battery approaches full charge.
- Charging is usually terminated when current falls below a cutoff (e.g. C/10).



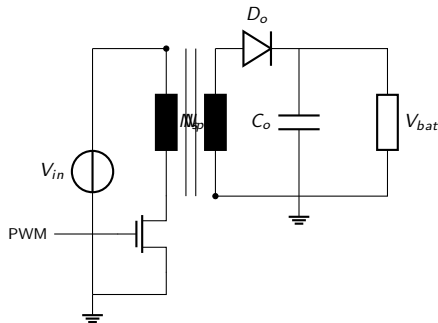
# Converter Choice: Flyback vs. Forward ?

Neither is “the” answer – it is a power-level and cost trade-off.

| Parameter           | Flyback                                     | Forward                                 |
|---------------------|---------------------------------------------|-----------------------------------------|
| Typical power range | Up to $\sim 150$ W                          | $\sim 100$ W to several hundred W       |
| Transformer         | Coupled inductor (stores energy in air gap) | True transformer, energy passes through |
| Output stage        | No separate output inductor needed          | Needs output inductor + freewheel diode |
| Component count     | Low – simple, cheap                         | Higher – more parts                     |
| Output ripple       | Higher                                      | Lower (continuous inductor current)     |

**For typical battery-charger power levels (a few W to  $\sim 150$  W), the flyback is the dominant choice**, which is why we develop it further here. Forward converters take over as power scales beyond that range.

# Flyback Converter – Circuit Diagram



Primary and secondary grounds are galvanically isolated. Dot placement (top of  $N_p$ , bottom of  $N_s$ ) is opposite – the defining feature of flyback: energy stores in  $N_p$  while the switch is ON, then transfers to the secondary and battery while the switch is OFF. Feedback to the PWM controller is normally routed back through an optocoupler to preserve isolation.

# Flyback – Design Equations & Worked Example

## 1) Steady-state voltage relation (CCM):

$$\frac{V_{out}}{V_{in}} = \frac{D}{1-D} n, \quad n = \frac{N_s}{N_p}, \quad D = \frac{V_{out}}{V_{in} n + V_{out}}$$

## 2) Worked example – charging a 12 V lead–acid battery from a 24 V DC bus:

Given:

➤  $V_{in} = 24 \text{ V}$ ,  $V_{out} = 14.4 \text{ V}$  (CV level)

➤  $n = N_s/N_p = 0.6$

➤  $I_{bat} = 3 \text{ A}$ ,  $\eta = 90\%$

➤  $f_{sw} = 100 \text{ kHz}$

$$D = \frac{14.4}{24(0.6) + 14.4} = 0.5$$

$$P_{out} = 14.4 \times 3 = 43.2 \text{ W}$$

$$P_{in} = 43.2/0.9 = 48 \text{ W}$$

$$I_{in,avg} = 48/24 = 2 \text{ A}$$

# What is a UPS, and Why It's Needed

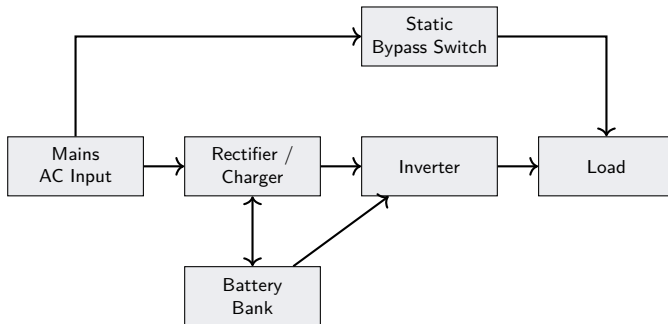
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- A UPS (Uninterruptible Power Supply) keeps a load powered through a mains failure, using a battery bank as the energy reserve.
- Protects against blackouts, sags, surges, and brownouts that a raw mains feed cannot guarantee against.
- Bridges the gap until a backup generator starts, or provides enough time for a controlled, safe shutdown.
- Essential for servers, medical equipment, industrial control systems, and other critical loads.

# UPS Types

| Type                       | How it Works                                                                                    | Typical Use                           |
|----------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------|
| Offline / Standby          | Load runs directly on mains; switches to battery+inverter on failure (~few ms transfer time)    | Basic desktop PCs, low-cost loads     |
| Line-Interactive           | Adds automatic voltage regulation (tap-changing) while on mains; switches to battery on failure | Small business, network equipment     |
| Online (Double-Conversion) | Continuously converts AC→DC→AC; load always runs from the inverter, zero transfer time          | Data centres, medical, critical loads |

# UPS – Block Diagram



The static bypass switch (online UPS) routes mains directly to the load for maintenance or if the inverter path fails, without dropping power.

# UPS – Backup Time & Worked Example

**Backup time equation:**

$$t_{\text{backup}} = \frac{\text{Battery capacity (Ah)} \times V_{\text{bat}} \times \text{DoD} \times \eta_{\text{inv}}}{P_{\text{load}}}$$

DoD: depth of discharge (safe fraction of capacity usable);  $\eta_{\text{inv}}$ : inverter efficiency.

**Worked example:**

**Given:**

- Battery: 100 Ah at 12 V
- DoD = 80 %
- $\eta_{\text{inv}} = 90 \%$
- $P_{\text{load}} = 200 \text{ W}$

$$E_{\text{avail}} = 100 \times 12 \times 0.8 = 960 \text{ Wh}$$

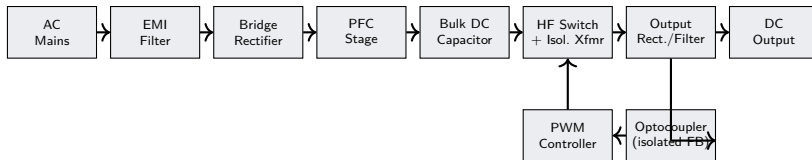
$$E_{\text{usable}} = 960 \times 0.9 = 864 \text{ Wh}$$

$$t_{\text{backup}} = \frac{864}{200} = 4.32 \text{ h}$$

# What is SMPS? Linear vs. Switching $\rightleftarrows$

| Aspect        | Linear Regulator                                  | Switching (SMPS) Regulator                    |
|---------------|---------------------------------------------------|-----------------------------------------------|
| Pass element  | Operates in linear (active) region                | Operates fully ON/OFF                         |
| Efficiency    | Low, $\sim 30\text{--}60\%$                       | High, $\sim 80\text{--}95\%$                  |
| Size / weight | Larger (line-frequency transformer, big heatsink) | Smaller (high-frequency transformer/inductor) |
| Noise         | Very low                                          | Switching noise / EMI present                 |
| Complexity    | Simple                                            | More complex (control loop, magnetics)        |

# Complete SMPS – Block Diagram



PFC (power-factor correction) is optional at low power but is mandatory in many regions above  $\sim 75$  W. The optocoupler preserves galvanic isolation while closing the feedback loop.

# SMPS Topology Classification

| Topology   | Isolated? | Typical Use                          |
|------------|-----------|--------------------------------------|
| Buck       | No        | Step-down, point-of-load regulation  |
| Boost      | No        | Step-up, PFC front-end               |
| Buck-Boost | No        | Step-up/down, small signal supplies  |
| Flyback    | Yes       | Low-medium power, chargers, adapters |
| Forward    | Yes       | Medium-high power                    |

This is exactly the flyback-vs-forward choice from the battery-charging section – the same trade-off applies across SMPS design generally.

# SMPS – Key Design Equations

1) Buck converter (step-down, ideal CCM):

$$V_{out} = D V_{in}$$

2) Boost converter (step-up, ideal CCM):

$$V_{out} = \frac{V_{in}}{1 - D}$$

3) Flyback (recap from earlier):

$$V_{out} = \frac{D}{1 - D} n V_{in}, \quad n = N_s / N_p$$

## Worked Example – Buck Output Filter Sizing ✂

### Given:

- $V_{in} = 12\text{ V}$ ,  $V_{out} = 5\text{ V}$
- $f_{sw} = 200\text{ kHz}$
- $I_{out} = 2\text{ A}$ , ripple  $\Delta I_L = 20\%$  of  $I_{out}$

### Design:

$$D = 5/12 = 0.417$$

$$\Delta I_L = 0.2 \times 2 = 0.4\text{ A}$$

$$L = \frac{(V_{in} - V_{out})D}{f_{sw}\Delta I_L} \approx 36.5\text{ }\mu\text{H}$$

**Selected:**  $L = 39\text{ }\mu\text{H}$  (nearest standard value)

Output Capacitor (ideal minimum, ripple  $\leq 50\text{ mV}$ )

$C_{out} \geq \frac{\Delta I_L}{8f_{sw}\Delta V_{out}} = 5\text{ }\mu\text{F}$  (minimum). In practice, ESR and transient response usually push the selected capacitor well above this – treat it as a floor, not a final value.

# Comparison: Charger vs. UPS vs. Generic SMPS

| Aspect            | Battery Charger                                    | UPS                           |
|-------------------|----------------------------------------------------|-------------------------------|
| Primary function  | Replenish battery energy                           | Backup power on mains failure |
| Typical topology  | Flyback (low-med power),<br>Forward (higher power) | Rectifier/charger + inverter  |
| Key design driver | CC-CV accuracy                                     | Backup time, transfer time    |
| Isolation         | Usually yes (safety)                               | Yes (inverter stage)          |

A generic SMPS sits underneath both: buck, boost, flyback, or forward, chosen by power level, isolation need, and application – the same design questions asked in every section of this session.

## Practical Tips & Summary ✓

- Choose flyback vs. forward (and buck vs. boost) by power level and isolation need first – component count and cost follow from that choice.
- CC–CV charging profiles protect both battery life and safety; always respect the chemistry's datasheet limits.
- UPS backup time is a direct energy-budget calculation – size the battery to the load, not the other way round.
- A complete SMPS is more than the switching stage: EMI filtering, PFC, and isolated feedback are just as essential to a working design.
- Treat every equation here as a starting point; validate on the bench before finalizing a design.

References: N. Mohan, T. M. Undeland, W. P. Robbins, *Power Electronics: Converters, Applications and Design*; M. H. Rashid, *Power Electronics: Circuits, Devices and Applications*.